

Module: Lost Found

pawguard/modules/lost_found — API prefix: /lost-found

PawGuard Backend | FastAPI 3.13 + PostgreSQL 17 + Redis/ARQ + S3 | Generated from repo @ commit 99a8430 (2026-08-12)

1. Purpose

Lost-pet and found-pet reporting with an automated weighted matching engine, GPS-based distance scoring, broadcast alerts, and an ownership-claim review workflow. See the dedicated Lost Pet Alert System document set for full detail.

2. Position in the Layered Architecture

Client → Router (auth+validate) → Service (business logic) → Repository (data access) → PostgreSQL

Like every module in this codebase, business rules live only in the Service layer; Routers authenticate/authorize/validate and Repositories perform data access with no business decisions (see AGENTS.md — the project's engineering constitution).

3. Data Model — Owned Tables

Model Class	Table Name
LostReport	lost_reports
FoundReport	found_reports
ReportMatch	report_matches

4. API Endpoints (17 total)

Method	Path	Permission Required
POST	/lost-found/lost	—
POST	/lost-found/lost/{report_id}/broadcast	lost_found:broadcast
POST	/lost-found/found	—
GET	/lost-found/lost	—
GET	/lost-found/found	—
GET	/lost-found/reunion-stories	—
GET	/lost-found/lost/{report_id}	—
GET	/lost-found/found/{report_id}	—
GET	/lost-found/lost/{report_id}/matches	—
GET	/lost-found/found/{report_id}/matches	—
POST	/lost-found/matches/{match_id}/claim	—
POST	/lost-found/matches/{match_id}/claim/review	system:admin
POST	/lost-found/matches/{match_id}/resolve	system:admin
DELETE	/lost-found/lost/{report_id}	system:admin
DELETE	/lost-found/found/{report_id}	system:admin
POST	/lost-found/lost/bulk/delete	system:admin
POST	/lost-found/found/bulk/delete	system:admin

Distinct permission codes used: lost_found:broadcast, system:admin

5. Key Service-Layer Operations

- bulk_delete_found_reports
- bulk_delete_lost_reports
- get_match
- get_matches_for_found
- get_matches_for_lost
- list_found_reports_paginated
- list_lost_reports_paginated
- list_matches_paginated
- queue_lost_alert_broadcast
- report_found_pet
- report_lost_pet
- resolve_found_report
- resolve_lost_report
- review_ownership_claim

+4 additional service methods.

6. Notable Implementation Details

- Weighted match scoring (breed 25 / color 25 / distance 20 / time 15 / collar 10 / markers 5) via Haversine distance.
- Broadcast is queued on ARQ (async, rate-limited 3/hour) rather than sent inline from the request.

Auto-extracted from pawguard/modules/lost_found/ (models.py, router(s), service.py) at repo commit 99a8430, 2026-08-12.